

Problem 1

Determine the speed at which a point in Belgrade moves due to the Earth's rotation around its own axis. The latitude of Belgrade is $44^\circ 45'$, and the radius of the Earth is approximately 6470 km.

Solution

The speed of a point on the surface of the Earth due to the Earth's rotation is calculated as the tangential speed at the given latitude. The formula for the tangential speed is:

$$v = \omega R \cos \varphi,$$

where:

- $\omega = \frac{2\pi}{86400 \text{ s}} \approx 7.27 \times 10^{-5} \text{ rad/s}$ is the angular velocity of the Earth (one full rotation in 24 hours),
- $R = 6470 \times 10^3 \text{ m}$ is the radius of the Earth,
- $\varphi = 44^\circ 45' = 44.75^\circ$ is the latitude of Belgrade.

Substituting the values into the formula:

$$v = (7.27 \times 10^{-5}) \times (6470 \times 10^3) \times \cos(44.75^\circ).$$

Evaluating the cosine term:

$$\cos(44.75^\circ) \approx 0.707.$$

Substituting this into the equation:

$$v \approx (7.27 \times 10^{-5}) \times (6470 \times 10^3) \times 0.707.$$

Carrying out the multiplication:

$$v \approx 7.27 \times 10^{-5} \times 6470000 \times 0.707 \approx 332.6 \text{ m/s}.$$

Final Answer

The speed of a point in Belgrade due to the Earth's rotation is approximately 332.6 m/s.